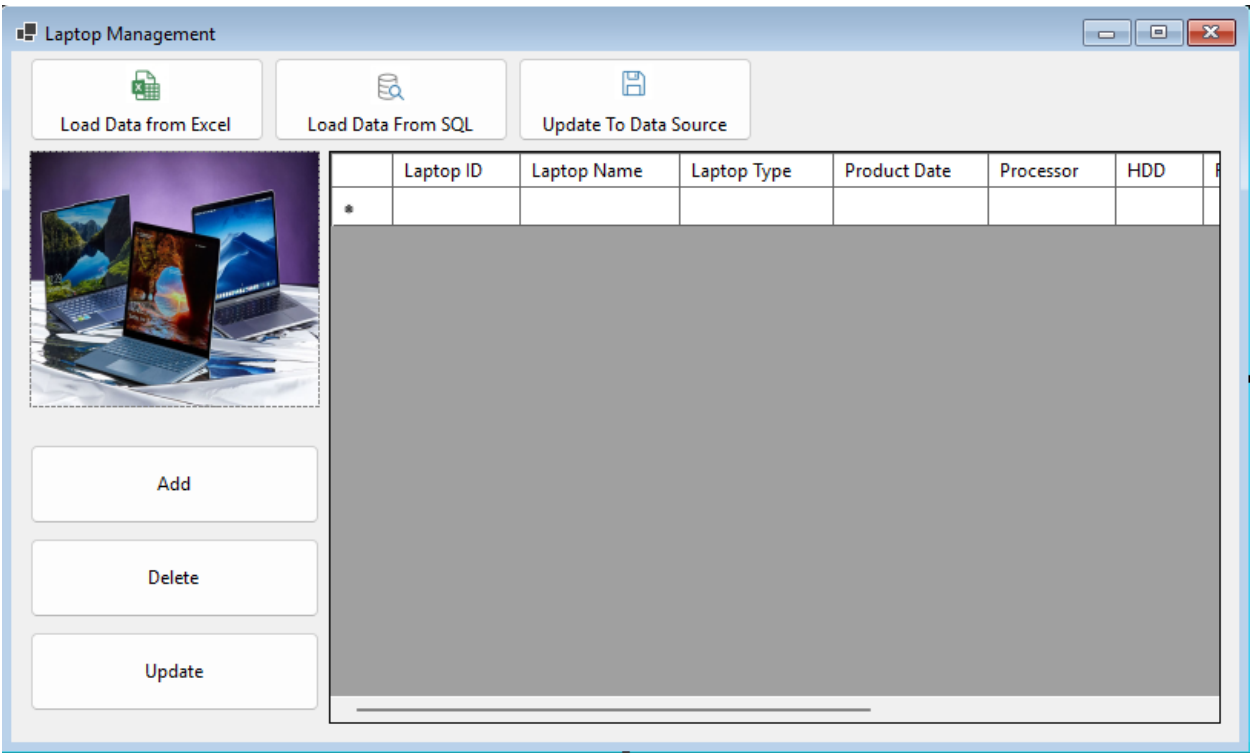


Bài Thực Hành Tuần 5 (LTNet)

1/ Thiết kế giao diện



2/ Thiết kế cơ sở dữ liệu

File Excel : LaptopDB.xlsx

	A	B	C	D	E	F	G	H	I	Formula Bar	K	L	M	N
1	LaptopID	LaptopName	LaptopType	ProductDate	Processor	HDD	RAM	Price	ImageName	Color	Weight			
2	LP001	Dell XPS 13 Plus	Ultrabook	01/01/2026	Core i7-1360P	512GB SSD	16GB	45000000	dell_xps_13.jpg	Platinum	1.23kg			
3	LP002	MacBook Air M2	Ultrabook	01/01/2026	Apple M2	256GB SSD	8GB	38000000	macbook_air_m2.jpg	Midnight	1.24kg			
4	LP003	Asus ROG Strix G15	Gaming	01/01/2026	Ryzen 7 6800H	1TB SSD	16GB	32000000	rog_strix_g15.jpg	Eclipse Gray	2.30kg			
5	LP004	HP Spectre x360	2-in-1	01/01/2026	Core i7-1255U	1TB SSD	16GB	42000000	hp_spectre.jpg	Nightfall Black	1.35kg			
6	LP005	Lenovo Legion 5	Gaming	01/01/2026	Core i5-12500H	512GB SSD	16GB	29000000	lenovo_legion5.jpg	Storm Grey	2.40kg			
7	LP006	Acer Swift 5	Văn phòng	01/01/2026	Core i7-1260P	512GB SSD	16GB	25000000	acer_swift5.jpg	Mist Green	1.20kg			
8	LP007	MSI Raider GE78	Gaming	01/01/2026	Core i9-13980HX	2TB SSD	32GB	89000000	msi_raider.jpg	Core Black	3.10kg			
9	LP008	MacBook Pro 14	Đồ họa	01/01/2026	Apple M2 Pro	512GB SSD	16GB	49000000	macbook_pro14.jpg	Space Gray	1.60kg			
10	LP009	LG Gram 17	Văn phòng	01/01/2026	Core i7-1360P	1TB SSD	16GB	39000000	lg_gram_17.jpg	White	1.35kg			
11	LP010	Surface Laptop 5	Văn phòng	01/01/2026	Core i5-1235U	256GB SSD	8GB	24000000	surface_lap5.jpg	Sage	1.27kg			
12	LP011	Dell Inspiron 15	Sinh viên	01/01/2026	Core i5-1135G7	512GB SSD	8GB	15000000	dell_inspiron.jpg	Silver	1.85kg			
13	LP012	Acer Nitro 5	Gaming	01/01/2026	Core i5-12500H	512GB SSD	8GB	19000000	acer_nitro5.jpg	Obsidian Black	2.50kg			
14	LP013	ThinkPad X1 Carbon	Doanh nhân	01/01/2026	Core i7-1355U	1TB SSD	32GB	55000000	thinkpad_x1.jpg	Deep Black	1.12kg			
15	LP014	Asus ZenBook 14	Văn phòng	01/01/2026	Core i5-1240P	512GB SSD	8GB	21000000	zenbook_14.jpg	Ponder Blue	1.39kg			
16	LP015	HP Victus 16	Gaming	01/01/2026	Ryzen 5 6600H	512GB SSD	16GB	22000000	hp_victus.jpg	Mica Silver	2.46kg			
17	LP016	Lenovo Yoga 9i	2-in-1	01/01/2026	Core i7-1360P	1TB SSD	16GB	46000000	yoga_9i.jpg	Oatmeal	1.40kg			
18	LP017	Razer Blade 15	Gaming	01/01/2026	Core i7-12800H	1TB SSD	32GB	65000000	razer_blade15.jpg	Matte Black	2.01kg			
19	LP018	Samsung Galaxy Book3	Văn phòng	01/01/2026	Core i7-1360P	512GB SSD	16GB	35000000	samsung_book3.jpg	Graphite	1.56kg			
20	LP019	MSI Modern 14	Sinh viên	01/01/2026	Core i3-1115G4	256GB SSD	8GB	11000000	msi_modern14.jpg	Classic Black	1.30kg			
21	LP020	Alienware x15 R2	Gaming	01/01/2026	Core i9-12900H	2TB SSD	32GB	95000000	alienware_x15.jpg	Lunar Light	2.34kg			

Database LaptopDB trong SQL

Max Rows: 1000										
	LaptopID	LaptopName	LaptopType	ProductDate	Processor	HDD	RAM	Price	ImageName	
	LP001	Dell XPS 13 Plus...	Ultrabook	5/15/2023 12:00...	Core i7-1360P ...	512GB SSD	16GB	45000000	dell_xps_13...	
	LP002	MacBook Air M...	Ultrabook	7/20/2022 12:00...	Apple M2	256GB SSD	8GB	28000000	macbook_a...	
	LP003	Asus ROG Strix ...	Gaming	3/10/2023 12:00...	Ryzen 7 6800H ...	1TB SSD	16GB	32000000	rog_strix_g...	
	LP004	HP Spectre x36...	2-in-1	1/5/2023 12:00:...	Core i7-1255U ...	1TB SSD	16GB	42000000	hp_spectre...	
	LP005	Lenovo Legion ...	Gaming	6/12/2023 12:00...	Core i5-12500H...	512GB SSD	16GB	29000000	lenovo_legi...	
	LP006	Acer Swift 5 ...	Văn phòng	11/20/2022 12:0...	Core i7-1260P ...	512GB SSD	16GB	25000000	acer_swift5...	
	LP007	MSI Raider GE7...	Gaming	8/1/2023 12:00:...	Core i9-13980H...	2TB SSD	32GB	89000000	msi_raider.j...	
	LP008	MacBook Pro 1 ...	Đồ họa	2/15/2023 12:00...	Apple M2 Pro ...	512GB SSD	16GB	49000000	macbook_p...	
	LP009	LG Gram 17 ...	Văn phòng	4/25/2023 12:00...	Core i7-1360P ...	1TB SSD	16GB	39000000	lg_gram_17...	
	LP010	Surface Laptop ...	Văn phòng	10/25/2022 12:0...	Core i5-1235U ...	256GB SSD	8GB	24000000	surface_lap...	
	LP011	Dell Inspiron 15...	Sinh viên	1/10/2023 12:00...	Core i5-1135G7 ...	512GB SSD	8GB	15000000	dell_inspiro...	
	LP012	Acer Nitro 5 ...	Gaming	12/5/2022 12:00...	Core i5-12500H...	512GB SSD	8GB	19000000	acer_nitro5...	
	LP013	ThinkPad X1 Ca...	Doanh nhân ...	3/30/2023 12:00...	Core i7-1355U ...	1TB SSD	32GB	55000000	thinkpad_x1...	
	LP014	Asus ZenBook ...	Văn phòng	5/5/2023 12:00:...	Core i5-1240P ...	512GB SSD	8GB	21000000	zenbook_14...	
	LP015	HP Victus 16 ...	Gaming	2/28/2023 12:00...	Ryzen 5 6600H ...	512GB SSD	16GB	22000000	hp_victus.jp...	
	LP016	Lenovo Yoga 9i ...	2-in-1	4/15/2023 12:00...	Core i7-1360P ...	1TB SSD	16GB	46000000	yoga_9i.jpg	
	LP017	Razer Blade 15 ...	Gaming	1/20/2023 12:00...	Core i7-12800H...	1TB SSD	32GB	65000000	razer_blade...	
	LP018	Samsung Galax...	Văn phòng	2/1/2023 12:00:...	Core i7-1360P ...	512GB SSD	16GB	35000000	samsung_b...	
	LP019	MSI Modern 14 ...	Sinh viên	9/15/2022 12:00...	Core i3-1115G4 ...	256GB SSD	8GB	11000000	msi_modern...	
	LP020	Alienware x15 R...	Gaming	11/30/2022 12:0...	Core i9-12900H...	2TB SSD	32GB	95000000	alienware_x...	
	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	
Connection Ready (localdb)\\MSSQLLocalDB LAPTOP-PABU7CR8\\Tam Ng... LaptopDB										

3/ Khai báo các cấu trúc dữ liệu sẽ sử dụng trong chương trình

Laptop.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Reflection;
5 using System.Text;
6 using System.Threading.Tasks;
7
8 namespace LaptopManagement
9 {
10     14 references
11     public class Laptop
12     {
13         11 references
14         public string LaptopID { get; set; }
15         10 references
16         public string LaptopName { get; set; }
17         9 references
18         public string LaptopType { get; set; }
19         10 references
20         public DateTime ProductDate { get; set; }
21         9 references
22         public string Processor { get; set; }
23         9 references
24         public string HDD { get; set; }
25         9 references
26         public string RAM { get; set; }
27         9 references
28         public int Price { get; set; }
29         10 references
30         public string ImageName { get; set; }
31         9 references
32         public string Color { get; set; }
33         9 references
34         public string Weight { get; set; }
35         3 references
36         public Laptop()
37         {
38             LaptopID = "Not Assigned";
39             LaptopName = "Not Assigned";
40         }
41     }
42 }
```

Form1.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Data;
5 using System.Drawing;
6 using System.Linq;
7 using System.Text;
8 using System.Threading.Tasks;
9 using System.Windows.Forms;
10 using System.IO;
11 using Excel = Microsoft.Office.Interop.Excel;
12 using System.Globalization;
13 using System.Data.SqlClient;
14
15 namespace LaptopManagement
16 {
17     4 references
18     public partial class Form1 : Form
19     {
20         1 reference
21         public Form1()
22         {
23             InitializeComponent();
24         }
25         public List<Laptop> LTLIST = new List<Laptop>();
26         // loadData = 0 (chưa có dữ liệu);
27         // loadData = 1 (có dữ liệu từ excel);
28         // loadData = 2 (có dữ liệu từ sql);
29         public int loadData = 0;
30         static string ProjectPath = Directory.GetParent(Directory.GetCurrentDirectory()).Parent.FullName;
31         string ExcelFilePath = ProjectPath + "\\Data\\LaptopDB.xlsx";
32         string connectionString = @"Data Source=(localdb)\MSSQLLocalDB;
33             Initial Catalog=LaptopDB; Integrated Security=True";
34         int CurrentLaptopIndex = -1;
35         DataTable datatable;
36         BindingSource binding = new BindingSource();
37     }
38 }
```

4/ Kết nối và lấy dữ liệu từ Excel File, SQL Server

```
71     DataRow newrow;
72     foreach (var bi in sublist)
73     {
74         newrow = datatable.NewRow();
75         newrow["LaptopID"] = bi.LaptopID;
76         newrow["LaptopName"] = bi.LaptopName;
77         newrow["LaptopType"] = bi.LaptopType;
78         newrow["ProductDate"] = bi.ProductDate;
79         newrow["Processor"] = bi.Processor;
80         newrow["HDD"] = bi.HDD;
81         newrow["RAM"] = bi.RAM;
82         newrow["Price"] = bi.Price;
83         newrow["ImageName"] = bi.ImageName;
84         newrow["Color"] = bi.Color;
85         newrow["Weight"] = bi.Weight;
86
87         datatable.Rows.Add(newrow);
88         datatable.AcceptChanges();
89     }
90
91     binding.AllowNew = true;
92     binding.DataSource = datatable;
93
94     dgwLaptopList.AutoGenerateColumns = false;
95     dgwLaptopList.DataSource = binding;
96 }
97
98 1 reference
99 public int ReadDataFromFile(List<Laptop> DataList, string FilePath, int colCount)
100 {
101     Excel.Application xlApp = new Excel.Application();
102     Excel.Workbook xlWorkbook = xlApp.Workbooks.Open(FilePath);
103     Excel.Worksheet xlWorksheet = xlWorkbook.Sheets[1];
104     Excel.Range xlRange = xlWorksheet.UsedRange;
105
106     xlWorksheet.Columns.ClearFormats();
107     xlWorksheet.Rows.ClearFormats();
108
109     int rowCount = xlWorksheet.UsedRange.Rows.Count;
110
111     int numLaptop = 0;
```

```
109
110     int numLaptop = 0;
111     string LaptopID = "";
112     string LaptopName = "";
113     string LaptopType = "";
114     DateTime ProductDate = DateTime.Now;
115     string Processor = "";
116     string HDD = "";
117     string RAM = "";
118     int Price = 0;
119     string ImageName = "";
120     string Color = "";
121     string Weight = "";
122
123     for (int i = 2; i <= rowCount; i++)
124     {
125         for (int j = 1; j <= colCount; j++)
126         {
127             switch (j)
128             {
129                 case 1: // LaptopID
130                     LaptopID = xlRange.Cells[i, j].Value2.ToString();
131                     break;
132                 case 2: // LaptopName
133                     LaptopName = xlRange.Cells[i, j].Value2.ToString();
134                     break;
135                 case 3: // LaptopType
136                     LaptopType = xlRange.Cells[i, j].Value2.ToString();
137                     break;
138                 case 4: // ProductDate
139                     try
140                     {
141                         ProductDate = DateTime.ParseExact(xlRange.Cells[i, j].Value2.ToString(),
142                             "dd/MM/yyyy", CultureInfo.InvariantCulture);
143                     }
144                     catch { ProductDate = DateTime.Now; }
145                     break;
146                 case 5: // Processor
147                     Processor = xlRange.Cells[i, j].Value2.ToString();
148                     break;
149                 case 6: // HDD
```

```

149         case 6: // HDD
150             HDD = xlRange.Cells[1, j].Value2.ToString();
151             break;
152         case 7: // RAM
153             RAM = xlRange.Cells[1, j].Value2.ToString();
154             break;
155         case 8: // Price
156             Price = Convert.ToInt32(xlRange.Cells[1, j].Value2.ToString());
157             break;
158         case 9: // ImageName
159             ImageName = xlRange.Cells[1, j].Value2.ToString();
160             break;
161         case 10: // Color
162             Color = xlRange.Cells[1, j].Value2.ToString();
163             break;
164         case 11: // Weight
165             Weight = xlRange.Cells[1, j].Value2.ToString();
166             break;
167     }
168 }
169 DataList.Add(new Laptop());
170 DataList[numLaptop].LaptopID = LaptopID;
171 DataList[numLaptop].LaptopName = LaptopName;
172 DataList[numLaptop].LaptopType = LaptopType;
173 DataList[numLaptop].ProductDate = ProductDate;
174 DataList[numLaptop].Processor = Processor;
175 DataList[numLaptop].HDD = HDD;
176 DataList[numLaptop].RAM = RAM;
177 DataList[numLaptop].Price = Price;
178 DataList[numLaptop].ImageName = ImageName;
179 DataList[numLaptop].Color = Color;
180 DataList[numLaptop].Weight = Weight;
181 numLaptop = numLaptop + 1;
182 }
183 xlApp.Quit();
184
185 MessageBox.Show("Load Data From Excel Finished! : " + (rowCount - 1).ToString() + " Records");
186
187 return (rowCount - 1);
188 }

```

```

189 1 reference
190 private void dgwLaptopList_SelectionChanged(object sender, EventArgs e)
191 {
192     if (LTList.Count == 0 || datatable.Rows.Count == 0 || dgwLaptopList.CurrentRow == null)
193         return;
194
195     CurrentLaptopIndex = dgwLaptopList.CurrentRow.Index;
196
197     if (CurrentLaptopIndex >= 0 && CurrentLaptopIndex < LTList.Count)
198     {
199         string imageName = LTList[CurrentLaptopIndex].ImageName;
200         string baseDir = AppDomain.CurrentDomain.BaseDirectory;
201         string projectPath = Directory.GetParent(Directory.GetParent(Directory.GetParent(baseDir).FullName).FullName).FullName;
202         string fullPath = Path.Combine(projectPath, "Data", imageName);
203
204         if (File.Exists(fullPath))
205         {
206             picLaptopImage.Image = Image.FromFile(fullPath);
207         }
208         else
209         {
210             picLaptopImage.Image = null;
211         }
212     }
213 }
214 1 reference
215 private void btnLoadSQL_Click(object sender, EventArgs e)
216 {
217     loadData = 2;
218     datatable = new DataTable();
219     LTList.Clear();
220
221     int NumDataRow = ReadDataFromSQLServer(LTList, connectionString);

```

```

221
222     var sublist = LTList.Select(x => new
223     {
224         LaptopID = x.LaptopID,
225         LaptopName = x.LaptopName,
226         LaptopType = x.LaptopType,
227         ProductDate = x.ProductDate.ToString("dd/MM/yyyy"),
228         Processor = x.Processor,
229         HDD = x.HDD,
230         RAM = x.RAM,
231         Price = x.Price.ToString() + " VND",
232         ImageName = x.ImageName,
233         Color = x.Color,
234         Weight = x.Weight
235     }).ToList();
236
237     datatable.Columns.Add("LaptopID");
238     datatable.Columns.Add("LaptopName");
239     datatable.Columns.Add("LaptopType");
240     datatable.Columns.Add("ProductDate");
241     datatable.Columns.Add("Processor");
242     datatable.Columns.Add("HDD");
243     datatable.Columns.Add("RAM");
244     datatable.Columns.Add("Price");
245     datatable.Columns.Add("ImageName");
246     datatable.Columns.Add("Color");
247     datatable.Columns.Add("Weight");
248
249     DataRow newrow;
250     foreach (var bi in sublist)
251     {

```

```

252         newrow = datatable.NewRow();
253         newrow["LaptopID"] = bi.LaptopID;
254         newrow["LaptopName"] = bi.LaptopName;
255         newrow["LaptopType"] = bi.LaptopType;
256         newrow["ProductDate"] = bi.ProductDate;
257         newrow["Processor"] = bi.Processor;
258         newrow["HDD"] = bi.HDD;
259         newrow["RAM"] = bi.RAM;
260         newrow["Price"] = bi.Price;
261         newrow["ImageName"] = bi.ImageName;
262         newrow["Color"] = bi.Color;
263         newrow["Weight"] = bi.Weight;
264
265         datatable.Rows.Add(newrow);
266         datatable.AcceptChanges();
267     }
268
269     binding.AllowNew = true;
270     binding.DataSource = datatable;
271
272     dgwLaptopList.AutoGenerateColumns = false;
273     dgwLaptopList.DataSource = binding;
274 }
275
276 1 reference
277 public int ReadDataFromSQLServer(List<Laptop> DataList, string connectionString)
278 {
279     SqlConnection cnn;
280     cnn = new SqlConnection(connectionString);

```

```

280     int iRow = 0;
281     int NumRecords = 0;
282
283     try
284     {
285         cnn.Open();
286         string SqlString = @"SELECT
287             LaptopID,
288             LaptopName,
289             LaptopType,
290             ProductDate = Convert(varchar(10), ProductDate, 103),
291             Processor,
292             HDD,
293             RAM,
294             Price,
295             ImageName,
296             Color,
297             Weight
298             FROM dbo.Laptop";
299
300         using (var command = new SqlCommand(SqlString, cnn))
301         {
302             using (var reader = command.ExecuteReader())
303             {
304                 while (reader.Read())
305                 {
306                     DataList.Add(new Laptop());
307
308                     DataList[iRow].LaptopID = reader.GetString(0);
309                     DataList[iRow].LaptopName = reader.GetString(1);
310                     DataList[iRow].LaptopType = reader.GetString(2);

```

```

311
312         try
313         {
314             DataList[iRow].ProductDate = DateTime.ParseExact(reader.GetString(3),
315                 "dd/MM/yyyy",
316                 System.Globalization.CultureInfo.InvariantCulture);
317         }
318         catch
319         {
320             DataList[iRow].ProductDate = DateTime.Now;
321         }
322
323         DataList[iRow].Processor = reader.GetString(4);
324         DataList[iRow].HDD = reader.GetString(5);
325         DataList[iRow].RAM = reader.GetString(6);
326
327         DataList[iRow].Price = reader.GetInt32(7);
328
329         DataList[iRow].ImageName = reader.GetString(8);
330         DataList[iRow].Color = reader.GetString(9);
331         DataList[iRow].Weight = reader.GetString(10);
332
333         iRow = iRow + 1;
334     }
335 }
336
337 SqlCommand cmd = new SqlCommand("select count(*) from Laptop", cnn);
338 object result = cmd.ExecuteScalar();
339 NumRecords = int.Parse(result.ToString());
340
341 MessageBox.Show("Finish Load Data From SQL: " + NumRecords.ToString() + " Records");
342 cnn.Close();
343
344 catch (SQLException ex)
345 {
346     MessageBox.Show("Can not open connection ! : " + ex.Message);
347 }
348
349 return NumRecords;
350 }

```

5/ Thêm ràng buộc khi người dùng Edit lại thông tin trong DataGridView

```
351 private void dgwLaptopList_EditingControlShowing(object sender, DataGridViewEditingControlShowingEventArgs e)
352 {
353     if (dgwLaptopList.CurrentCell.ColumnIndex == 7)
354     {
355         TextBox tb = e.Control as TextBox;
356         if (tb != null)
357         {
358             tb.KeyPress -= new KeyPressEventHandler(ColumnPrice_KeyPress);
359             tb.KeyPress += new KeyPressEventHandler(ColumnPrice_KeyPress);
360         }
361     }
362 }
363
364 2 references
365 private void ColumnPrice_KeyPress(object sender, KeyPressEventArgs e)
366 {
367     if (!char.IsControl(e.KeyChar) && !char.IsDigit(e.KeyChar))
368     {
369         e.Handled = true;
370     }
371 }
```

6/ Thực hiện các nút nhấn Thêm, Xóa , Sửa, Cập Nhật

```
371 private void btnAdd_Click(object sender, EventArgs e)
372 {
373     Laptop lap = new Laptop();
374     lap.LaptopID = "Not Assigned";
375     lap.LaptopName = "Not Assigned";
376     lap.LaptopType = "Not Assigned";
377     lap.ProductDate = DateTime.ParseExact("01/01/1900", "dd/MM/yyyy", CultureInfo.InvariantCulture);
378     lap.Processor = "Not Assigned";
379     lap.HDD = "Not Assigned";
380     lap.RAM = "Not Assigned";
381     lap.Price = 0;
382     lap.ImageName = "laptop.jpg";
383     lap.Color = "Not Assigned";
384     lap.Weight = "Not Assigned";
385     LTList.Add(lap);
386
387     DataRow newrow;
388     newrow = datatable.NewRow();
389     newrow["LaptopID"] = lap.LaptopID;
390     newrow["LaptopName"] = lap.LaptopName;
391     newrow["LaptopType"] = lap.LaptopType;
392     newrow["ProductDate"] = lap.ProductDate.ToString("dd/MM/yyyy");
393     newrow["Processor"] = lap.Processor;
394     newrow["HDD"] = lap.HDD;
395     newrow["RAM"] = lap.RAM;
396     newrow["Price"] = lap.Price.ToString() + " VND";
397     newrow["ImageName"] = lap.ImageName;
398     newrow["Color"] = lap.Color;
399     newrow["Weight"] = lap.Weight;
400 }
```

```

401     datatable.Rows.Add(newrow);
402     datatable.AcceptChanges();
403
404     MessageBox.Show("Finish Adding");
405 }
406
407 1 reference
408 private void btnDelete_Click(object sender, EventArgs e)
409 {
410     Laptop lap;
411     if (CurrentLaptopIndex >= 0)
412         lap = LTList[CurrentLaptopIndex];
413     else
414         return;
415
416     string question = "Do You Want to delete Laptop: " + lap.LaptopID;
417     DialogResult result = MessageBox.Show(question, "Delete", MessageBoxButtons.YesNo, MessageBoxIcon.Question);
418
419     if (result == DialogResult.Yes)
420     {
421         LTList.RemoveAt(CurrentLaptopIndex);
422         binding.RemoveAt(CurrentLaptopIndex);
423     }
424     MessageBox.Show("Finish Delete");
425 }
426
427 1 reference
428 private void btnUpdate_Click(object sender, EventArgs e)
429 {
430     DataRow row;
431     for (int i = 0; i < datatable.Rows.Count; i++)
432     {

```

```

431         row = datatable.Rows[i];
432
433         LTList[i].LaptopID = row["LaptopID"].ToString();
434         LTList[i].LaptopName = row["LaptopName"].ToString();
435         LTList[i].LaptopType = row["LaptopType"].ToString();
436         LTList[i].ProductDate = DateTime.ParseExact(row["ProductDate"].ToString(), "dd/MM/yyyy", CultureInfo.InvariantCulture);
437         LTList[i].Processor = row["Processor"].ToString();
438         LTList[i].HDD = row["HDD"].ToString();
439         LTList[i].RAM = row["RAM"].ToString();
440
441         string sPrice = row["Price"].ToString();
442         LTList[i].Price = Convert.ToInt32(sPrice.Substring(0, sPrice.IndexOf(" ")));
443
444         LTList[i].ImageName = row["ImageName"].ToString();
445         LTList[i].Color = row["Color"].ToString();
446         LTList[i].Weight = row["Weight"].ToString();
447     }
448     MessageBox.Show("Finish Update");
449 }
450 1 reference
451 private void btnUpdateSource_Click(object sender, EventArgs e)
452 {
453     if (loadData == 1)
454         WriteDataToExcelFile(LTList, ExcelFilePath);
455     else
456         WriteDataToSQLServer(LTList, connectionString);
457 }
458 1 reference
459 public void WriteDataToExcelFile(List<Laptop> DataList, string ExcelFilePath)
460 {
461     Excel.Application xlApp = new Excel.Application();
462     Excel.Workbook xlWorkbook = xlApp.Workbooks.Open(ExcelFilePath);
463     Excel._Worksheet xlWorksheet = xlWorkbook.Sheets[1];

```

```

463 Excel.Range xlRange;
464 string[,] Data = new string[1, 11];
465
466 int idxRow = 2;
467 foreach (Laptop lap in DataList)
468 {
469     Data[0, 0] = lap.LaptopID;
470     Data[0, 1] = lap.LaptopName;
471     Data[0, 2] = lap.LaptopType;
472     Data[0, 3] = lap.ProductDate.ToString("dd/MM/yyyy", CultureInfo.InvariantCulture);
473     Data[0, 4] = lap.Processor;
474     Data[0, 5] = lap.HDD;
475     Data[0, 6] = lap.RAM;
476     Data[0, 7] = lap.Price.ToString();
477     Data[0, 8] = lap.ImageName;
478     Data[0, 9] = lap.Color;
479     Data[0, 10] = lap.Weight;
480
481     xlRange = xlWorksheet.get_Range("A" + idxRow.ToString(), "K" + idxRow.ToString());
482     xlRange.Value2 = Data;
483
484     idxRow = idxRow + 1;
485 }
486
487 xlWorkbook.Save();
488 xlWorkbook.Close();
489 xlApp.Quit();
490
491 MessageBox.Show("Finish Update to DataSource Excel");
492 }

```

```

493 public void WriteDataToSqlServer(List<Laptop> DataList, string connectionString)
494 {
495     SqlConnection cnn;
496     SqlCommand myCommand = new SqlCommand();
497     string query;
498
499     cnn = new SqlConnection(connectionString);
500     try
501     {
502         cnn.Open();
503
504         query = "TRUNCATE TABLE Laptop";
505         myCommand.CommandText = query;
506         myCommand.Connection = cnn;
507         myCommand.ExecuteNonQuery();
508
509         query = @"INSERT INTO Laptop(LaptopID, LaptopName, LaptopType,
510 | ProductDate, Processor, HDD, RAM, Price, ImageName, Color, Weight)";
511         query += @"VALUES (@LaptopID, @LaptopName, @LaptopType, @ProductDate,
512 | @Processor, @HDD, @RAM, @Price, @ImageName, @Color, @Weight)";
513
514         myCommand.CommandText = query;
515         myCommand.Connection = cnn;
516
517         myCommand.Parameters.Add(new SqlParameter("@LaptopID", SqlDbType.NVarChar));
518         myCommand.Parameters.Add(new SqlParameter("@LaptopName", SqlDbType.NVarChar));
519         myCommand.Parameters.Add(new SqlParameter("@LaptopType", SqlDbType.NVarChar));
520         myCommand.Parameters.Add(new SqlParameter("@ProductDate", SqlDbType.DateTime));
521         myCommand.Parameters.Add(new SqlParameter("@Processor", SqlDbType.NVarChar));
522         myCommand.Parameters.Add(new SqlParameter("@HDD", SqlDbType.NVarChar));
523         myCommand.Parameters.Add(new SqlParameter("@RAM", SqlDbType.NVarChar));
524         myCommand.Parameters.Add(new SqlParameter("@Price", SqlDbType.Int));
525         myCommand.Parameters.Add(new SqlParameter("@ImageName", SqlDbType.NVarChar));

```

```

526         myCommand.Parameters.Add(new SqlParameter("@Color", SqlDbType.NVarChar));
527         myCommand.Parameters.Add(new SqlParameter("@Weight", SqlDbType.NVarChar));
528
529         foreach (Laptop lap in DataList)
530         {
531             myCommand.Parameters[0].Value = lap.LaptopID;
532             myCommand.Parameters[1].Value = lap.LaptopName;
533             myCommand.Parameters[2].Value = lap.LaptopType;
534             myCommand.Parameters[3].Value = lap.ProductDate.ToString("yyyy-MM-dd", CultureInfo.InvariantCulture);
535             myCommand.Parameters[4].Value = lap.Processor;
536             myCommand.Parameters[5].Value = lap.HDD;
537             myCommand.Parameters[6].Value = lap.RAM;
538             myCommand.Parameters[7].Value = lap.Price;
539             myCommand.Parameters[8].Value = lap.ImageName;
540             myCommand.Parameters[9].Value = lap.Color;
541             myCommand.Parameters[10].Value = lap.Weight;
542
543             myCommand.ExecuteNonQuery();
544         }
545         cnn.Close();
546     }
547     catch (SqlException ex)
548     {
549         MessageBox.Show("Can not open connection ! " + ex.Message);
550     }
551
552     MessageBox.Show("Finish Update to DataSource SQL Server");
553 }
554
555 }
556
557

```

7/ Các thao tác thêm xóa sửa trên DataGridView và UpdateToSource cho cả Excel và SQL

Load Data from Excel → Add

Laptop Management

Load Data from Excel

Load Data From SQL

Update To Data Source



Add

Delete

Update

	Laptop ID	Laptop Name	Laptop Type	Product Date	Processor	
▶	LP001	Dell XPS 13 Plus	Ultrabook	01/01/2026	Core i7-1360P	5'
	LP002	MacBook Air M2	Ultrabook	01/01/2026	Apple M2	2'
	LP003	Asus ROG Strix G15	Gaming	01/01/2026	Ryzen 7 6800H	1'
	LP004	HP Spectre x360	2-in-1	01/01/2026	Core i7-1255U	1'
	LP005	Lenovo Legion 5	Gaming	01/01/2026	Core i5-12500H	5'
	LP006	Acer Swift 5	Văn phòng	01/01/2026	Core i7-1260P	5'
	LP007	MSI Raider GE78	Gaming	01/01/2026	Core i9-13980HX	2'
	LP008	MacBook Pro 14	Đồ họa	01/01/2026	Apple M2 Pro	5'
	LP009	LG Gram 17	Văn phòng	01/01/2026	Core i7-1360P	1'
	LP010	Surface Laptop 5	Văn phòng	01/01/2026	Core i5-1235U	2'
	LP011	Dell Inspiron 15	Sinh viên	01/01/2026	Core i5-1135G7	5'
	LP012	Acer Nitro 5	Gaming	01/01/2026	Core i5-12500H	5'
	LP013	ThinkPad X1 Carbon	Doanh nhân	01/01/2026	Core i7-1355U	1'

Update to Data Source

1	A	B	C	D	E	F	G	H	I	J	K	L	M	N
2	LP001	Dell XPS 13 Plus	Ultrabook	01/01/2026	Core i7-1360P	512GB SSD	16GB	45000000	dell_xps_13.jpg	Platinum	1.23kg			
3	LP002	MacBook Air M2	Ultrabook	01/01/2026	Apple M2	256GB SSD	8GB	28000000	macbook_air_m2.jpg	Midnight	1.24kg			
4	LP003	Asus ROG Strix G15	Gaming	01/01/2026	Ryzen 7 6800H	1TB SSD	16GB	32000000	rog_strix_g15.jpg	Eclipse Gray	2.30kg			
5	LP004	HP Spectre x360	2-in-1	01/01/2026	Core i7-1255U	1TB SSD	16GB	42000000	hp_spectre.jpg	Nightfall Black	1.35kg			
6	LP005	Lenovo Legion 5	Gaming	01/01/2026	Core i5-12500H	512GB SSD	16GB	29000000	lenovo_legion5.jpg	Storm Grey	2.40kg			
7	LP006	Acer Swift 5	Văn phòng	01/01/2026	Core i7-1260P	512GB SSD	16GB	25000000	acer_swift5.jpg	Mist Green	1.20kg			
8	LP007	MSI Raider GE78	Gaming	01/01/2026	Core i9-13980HX	2TB SSD	32GB	89000000	msi_raider.jpg	Core Black	3.10kg			
9	LP008	MacBook Pro 14	Đồ họa	01/01/2026	Apple M2 Pro	512GB SSD	16GB	49000000	macbook_pro14.jpg	Space Gray	1.60kg			
10	LP009	LG Gram 17	Văn phòng	01/01/2026	Core i7-1360P	1TB SSD	16GB	39000000	lg_gram_17.jpg	White	1.35kg			
11	LP010	Surface Laptop 5	Văn phòng	01/01/2026	Core i5-1235U	256GB SSD	8GB	24000000	surface_lap5.jpg	Sage	1.27kg			
12	LP011	Dell Inspiron 15	Sinh viên	01/01/2026	Core i5-1135G7	512GB SSD	8GB	15000000	dell_inspiron.jpg	Silver	1.85kg			
13	LP012	Acer Nitro 5	Gaming	01/01/2026	Core i5-12500H	512GB SSD	8GB	19000000	acer_nitro5.jpg	Obsidian Black	2.50kg			
14	LP013	ThinkPad X1 Carbon	Doanh nhân	01/01/2026	Core i7-1355U	1TB SSD	32GB	55000000	thinkpad_x1.jpg	Deep Black	1.12kg			
15	LP014	Asus ZenBook 14	Văn phòng	01/01/2026	Core i5-1240P	512GB SSD	8GB	21000000	zenbook_14.jpg	Ponder Blue	1.39kg			
16	LP015	HP Victus 16	Gaming	01/01/2026	Ryzen 5 6600H	512GB SSD	16GB	22000000	hp_victus.jpg	Mica Silver	2.46kg			
17	LP016	Lenovo Yoga 9i	2-in-1	01/01/2026	Core i7-1360P	1TB SSD	16GB	46000000	yoga_9i.jpg	Oatmeal	1.40kg			
18	LP017	Razer Blade 15	Gaming	01/01/2026	Core i7-12800H	1TB SSD	32GB	65000000	razer_blade15.jpg	Matte Black	2.01kg			
19	LP018	Samsung Galaxy Book3	Văn phòng	01/01/2026	Core i7-1360P	512GB SSD	16GB	35000000	samsung_book3.jpg	Graphite	1.56kg			
20	LP019	MSI Modern 14	Sinh viên	01/01/2026	Core i3-1115G4	256GB SSD	8GB	11000000	msi_modern14.jpg	Classic Black	1.30kg			
21	LP020	Alienware x15 R2	Gaming	01/01/2026	Core i9-12900H	2TB SSD	32GB	95000000	alienware_x15.jpg	Lunar Light	2.34kg			
22	Not Assigned	Not Assigned	Not Assigned	01/01/1900	Not Assigned	Not Assigned	Not Assi	0	laptop.jpg	Not Assigned	Not Assigned			
23														

Delete

Laptop Management

Load Data from Excel

Load Data From SQL

Update To Data Source



Add

Delete

Update

Laptop ID	Laptop Name	Laptop Type	Product Date	Processor	RAM
LP010	Surface Laptop 5	Văn phòng	01/01/2026	Core i5-1235U	256GB
LP011	Dell Inspiron 15	Sinh viên	01/01/2026	Core i5-1135G7	512GB
LP012	Acer Nitro 5	Game	01/01/2026	Core i5-12500H	512GB
LP013	ThinkPad X1	Business	01/01/2026	Core i7-1355U	16GB
LP014	Asus ZenBook 14	Văn phòng	01/01/2026	Core i5-1240P	512GB
LP015	HP Victus 16	Game	01/01/2026	Ryzen 5 6600H	512GB
LP016	Lenovo Yoga 9i	2-in-1	01/01/2026	Core i7-1360P	16GB
LP017	Razer Blade 15	Gaming	01/01/2026	Core i7-12800H	16GB
LP018	Samsung Galaxy Book3	Văn phòng	01/01/2026	Core i7-1360P	512GB
LP019	MSI Modern 14	Sinh viên	01/01/2026	Core i3-1115G4	256GB
LP020	Alienware x15 R2	Gaming	01/01/2026	Core i9-12900H	256GB
▶	Not Assigned	Not Assigned	01/01/1900	Not Assigned	Not Assigned
*					

Delete

Do You Want to delete Laptop: Not Assigned

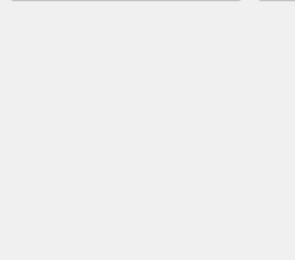
YesNo

Laptop Management

Load Data from Excel

Load Data From SQL

Update To Data Source



Add

Delete

Update

Laptop ID	Laptop Name	Laptop Type	Product Date	Processor	RAM	
LP009	LG Gram 17	Văn phòng	01/01/2026	Core i7-1360P	16GB	
LP010	Surface Laptop 5	Văn phòng	01/01/2026	Core i5-1235U	256GB	
LP011	Dell Inspiron 15	Sinh viên	01/01/2026	Core i5-1135G7	512GB	
LP012	Acer Nitro 5	Game	01/01/2026	Core i5-12500H	512GB	
LP013	ThinkPad X1 Carbon	Business	01/01/2026	Core i7-1355U	16GB	
LP014	Asus ZenBook 14	Văn phòng	01/01/2026	Core i5-1240P	512GB	
LP015	HP Victus 16	Game	01/01/2026	Ryzen 5 6600H	512GB	
LP016	Lenovo Yoga 9i	2-in-1	01/01/2026	Core i7-1360P	16GB	
LP017	Razer Blade 15	Gaming	01/01/2026	Core i7-12800H	16GB	
LP018	Samsung Galaxy Book3	Văn phòng	01/01/2026	Core i7-1360P	512GB	
LP019	MSI Modern 14	Sinh viên	01/01/2026	Core i3-1115G4	256GB	
▶	LP020	Alienware x15 R2	Gaming	01/01/2026	Core i9-12900H	256GB
*						

Finish Delete

OK

Load Data from SQL → Add → Update to DataSource

Laptop Management

Load Data from Excel

Load Data From SQL

Update To Data Source



Add

Delete

Update

	Laptop ID	Laptop Name	Laptop Type	Product Date	Processor	
	LP010	Surface Laptop 5	Văn phòng	01/01/2026	Core i5-1235U	2'
	LP011	Dell Inspiron 15	Sinh viên	01/01/2026	Core i5-1135G7	5'
	LP012	Acer Nitro 5	Gaming	01/01/2026	Core i5-12500H	5'
	LP013	ThinkPad X1 Carbon	Doanh nhân	01/01/2026	Core i7-1355U	1'
	LP014	Asus ZenBook 14	Văn phòng	01/01/2026	Core i5-1240P	5'
	LP015	HP Victus 16	Gaming	01/01/2026	Ryzen 5 6600H	5'
	LP016	Lenovo Yoga 9i	2-in-1	01/01/2026	Core i7-1360P	1'
	LP017	Razer Blade 15	Gaming	01/01/2026	Core i7-12800H	1'
	LP018	Samsung Galaxy Book3	Văn phòng	01/01/2026	Core i7-1360P	5'
	LP019	MSI Modern 14	Sinh viên	01/01/2026	Core i3-1115G4	2'
	LP020	Alienware x15 R2	Gaming	01/01/2026	Core i9-12900H	2'
▶	LP021	Not Assigned	Not Assigned	01/01/1900	Not Assigned	N
*						

Laptop Management

Load Data from Excel

Load Data From SQL

Update To Data Source



Add

Delete

Update

	Laptop ID	Laptop Name	Laptop Type	Product Date	Processor	
	LP010	Surface Laptop 5	Văn phòng	01/01/2026	Core i5-1235U	2'
	LP011	Dell Inspiron 15	Sinh viên	01/01/2026	Core i5-1135G7	5'
	LP012	Acer Nitro 5			Core i5-12500H	5'
	LP013	ThinkPad X1			Core i7-1355U	1'
	LP014	Asus ZenBoo			Core i5-1240P	5'
	LP015	HP Victus 16			Ryzen 5 6600H	5'
	LP016	Lenovo Yoga			Core i7-1360P	1'
	LP017	Razer Blade 15	Gaming	01/01/2026	Core i7-12800H	1'
	LP018	Samsung Galaxy Book3	Văn phòng	01/01/2026	Core i7-1360P	5'
	LP019	MSI Modern 14	Sinh viên	01/01/2026	Core i3-1115G4	2'
	LP020	Alienware x15 R2	Gaming	01/01/2026	Core i9-12900H	2'
▶	LP021	Not Assigned	Not Assigned	01/01/1900	Not Assigned	N
*						

Finish Update to DataSource SQL Server

OK

[illegible]